

$$2.) \nabla f(x, y) = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix}$$

$$2.1) f_1(x, y) = x^2 \cdot \sin(y)$$

↳ Gradient: part. Ableitung nach x : $y = \text{konstant}$

$$\frac{\partial f_1}{\partial x} = 2x \cdot \sin(y)$$

part. Ableitung nach y : $x = \text{konstant}$

$$\frac{\partial f_1}{\partial y} = x^2 \cdot \cos(y)$$

$$\Rightarrow \underline{\underline{\nabla f_1(x, y) = \begin{pmatrix} 2x \cdot \sin(y) \\ x^2 \cdot \cos(y) \end{pmatrix}}}$$

↳ Nullstellen: Gradient gleich Nullvektor

$$1: 2x \cdot \sin(y) = 0$$

$$2: x^2 \cdot \cos(y) = 0$$

$$\text{Fall 1: } x = 0 \Rightarrow 2 \cdot 0 \cdot \sin(y) = 0$$

↳ jedes $y \in \mathbb{R}$

$$\text{Fall 2: } \cos(y) = 0 \Rightarrow y = \frac{\pi}{2} + k \cdot \pi \quad | k \in \mathbb{Z}$$

$$\hookrightarrow \text{einsetzen: } 2x \cdot \sin\left(\frac{\pi}{2} + k \cdot \pi\right) \Rightarrow 1 \vee -1 \Rightarrow x = 0 \Rightarrow \text{Fall 1}$$

$$\left. \begin{array}{l} \text{Fall 1: } x = 0 \Rightarrow 2 \cdot 0 \cdot \sin(y) = 0 \\ \text{Fall 2: } \cos(y) = 0 \Rightarrow y = \frac{\pi}{2} + k \cdot \pi \quad | k \in \mathbb{Z} \\ \hookrightarrow \text{einsetzen: } 2x \cdot \sin\left(\frac{\pi}{2} + k \cdot \pi\right) \Rightarrow 1 \vee -1 \Rightarrow x = 0 \Rightarrow \text{Fall 1} \end{array} \right\} \underline{\underline{N_1 = \{(x, y) \in \mathbb{R} \mid x = 0, y \in \mathbb{R}\}}}$$

2.2)

$$\hookrightarrow f_2(x, y) = \sin^2(x) + y^{1/3}$$

↳ Gradienten

$$\hookrightarrow \text{nach } x: \frac{\partial f_2}{\partial x} = 2 \sin(x) \cdot \cos(x) = \sin(2x)$$

$$\hookrightarrow \text{nach } y: \frac{\partial f_2}{\partial y} = \frac{1}{3} \cdot y^{-2/3} = \frac{1}{3 \sqrt[3]{y^2}}$$

$$\left. \begin{array}{l} \frac{\partial f_2}{\partial x} = \sin(2x) \\ \frac{\partial f_2}{\partial y} = \frac{1}{3 \sqrt[3]{y^2}} \end{array} \right\} \underline{\underline{f_2(x, y) = \begin{pmatrix} \sin(2x) \\ \frac{1}{3 \sqrt[3]{y^2}} \end{pmatrix}}}$$

↳ Nullstellen

$$1. \sin(2x) = 0$$

$$2. \frac{1}{3 \sqrt[3]{y^2}} = 0 \Rightarrow \text{Fall 2 hat kein } y \in \mathbb{R}$$

$$\left. \begin{array}{l} 1. \sin(2x) = 0 \\ 2. \frac{1}{3 \sqrt[3]{y^2}} = 0 \Rightarrow \text{Fall 2 hat kein } y \in \mathbb{R} \end{array} \right\} \underline{\underline{N_2 = \emptyset}}$$

2.3)

$$f_3(x, y) = x \cdot e^{-(x^2 + y^2)}$$

↳ Gradient:

$$\left. \begin{array}{l} \hookrightarrow \frac{\partial f_3}{\partial x} = (1 - 2x^2) \cdot e^{-(x^2 + y^2)} \\ \hookrightarrow \frac{\partial f_3}{\partial y} = -2xy \cdot e^{-(x^2 + y^2)} \end{array} \right\} \underline{\underline{\nabla f_3(x, y) = \begin{pmatrix} (1 - 2x^2) \cdot e^{-(x^2 + y^2)} \\ -2xy \cdot e^{-(x^2 + y^2)} \end{pmatrix}}}$$

↳ Nullstellen:

$$\left. \begin{array}{l} 1. \quad 1 - 2x^2 = 0 \Rightarrow x^2 = \frac{1}{2} \Rightarrow x = \pm \frac{1}{\sqrt{2}} \\ 2. \quad -2xy = 0 \Rightarrow x = 0 \text{ oder } y = 0 \end{array} \right\} \underline{\underline{N_3 = \left\{ \left(\frac{1}{\sqrt{2}}, 0 \right), \left(-\frac{1}{\sqrt{2}}, 0 \right) \right\}}}$$

2.4)

$$f_4(x, y) = x^4 - 2x^2 + y^2$$

↳ Gradient

$$\left. \begin{array}{l} \hookrightarrow \frac{\partial f_4}{\partial x} = 4x^3 - 4x \\ \hookrightarrow \frac{\partial f_4}{\partial y} = 2y \end{array} \right\} \underline{\underline{\nabla f_4(x, y) = \begin{pmatrix} 4x^3 - 4x \\ 2y \end{pmatrix}}}$$

↳ Nullstellen

$$\left. \begin{array}{l} 1. \quad 4x^3 - 4x = 0 \Rightarrow 4x(x^2 - 1) = 0 \\ 2. \quad 2y = 0 \Rightarrow y = 0 \\ \hookrightarrow y = 0 \Rightarrow x_1 = 0, x_2 = 1, x_3 = -1 \end{array} \right\} \underline{\underline{N_4 = \{(0, 0), (1, 0), (-1, 0)\}}}$$

3)

$$a) \quad f_1(x) = \cos(x_1) \cdot \sin(x_2), \quad x \in \mathbb{R}^2$$

↳ Dimension: $x \in \mathbb{R}, f_1(x) \in \mathbb{R}$

$$\hookrightarrow \underline{\underline{\frac{\partial f_1}{\partial x} \in \mathbb{R}^{1 \times 2}}}$$

↳ Ableitung nach x_1 :

$$\frac{\partial f_1}{\partial x_1} = -\sin(x_1) \cdot \sin(x_2)$$

↳ Ableitung nach x_2 :

$$\frac{\partial f_1}{\partial x_2} = \cos(x_1) \cdot \cos(x_2)$$

$$\underline{\underline{\frac{\partial f_1}{\partial x} = (-\sin(x_1) \cdot \sin(x_2) \quad \cos(x_1) \cdot \cos(x_2))}}$$

3.1

b)

$$f_2(x, y) = x^T A y, \quad x, y \in \mathbb{R}^n \quad A \in \mathbb{R}^{n \times n}$$

↳ Dimension: $x \in \mathbb{R}^n$, Bilinearform ist ein Skalar

$$\Rightarrow \frac{\partial f_2}{\partial x} \in \mathbb{R}^{1 \times n}$$

↳ Jacobimatrix:

es gilt: $A y$ ist konstanter Vektor zu x
 $\Rightarrow x^T v$ mit $v = A y$

$$\hookrightarrow \frac{\partial (x^T v)}{\partial x} = v^T$$

$$\Rightarrow v^T = (A y)^T = y^T A^T$$

$$\frac{\partial f_2}{\partial x} = (A y)^T = y^T A^T$$

c) $f_3(x) = x^T x + \|x\|_1$ mit $x \in \mathbb{R}^n$

↳ Dimension: Da $x \in \mathbb{R}^n$ und $f_3(x) \in \mathbb{R}$ gilt unter der Bedingung $x_i \neq 0$.

$$\frac{\partial f_3}{\partial x} \in \mathbb{R}^{1 \times n}$$

↳ Jacobimatrix

1. Skalarprodukt = Summe der Quadrate

$$\hookrightarrow x^T x = \sum_{i=1}^n x_i^2$$

$$\hookrightarrow \frac{\partial (x^T x)}{\partial x} = 2x^T = (2x_1, 2x_2, \dots, 2x_n)$$

2. L_1 -Norm = Summe der Beträge

$$\hookrightarrow \|x\|_1 = \sum_{i=1}^n |x_i|$$

$$\hookrightarrow \frac{\partial \|x\|_1}{\partial x} = (\operatorname{sgn}(x_1), \operatorname{sgn}(x_2), \dots, \operatorname{sgn}(x_n))$$

$$\Rightarrow \frac{\partial f_3}{\partial x} = (2x_1 + \operatorname{sgn}(x_1), 2x_2 + \operatorname{sgn}(x_2), \dots, 2x_n + \operatorname{sgn}(x_n))$$

5.)

1.) $f_3(z) = \sin(z)$ mit $z = Ax + b$; $\frac{df}{dx} = \frac{df}{dz} \cdot \frac{dz}{dx}$

$\hookrightarrow A \in \mathbb{R}^{m \times n}, x \in \mathbb{R}^n, b \in \mathbb{R}^m$

$\hookrightarrow$ Innere Ableitung

$$\frac{dz}{dx} = A \in \mathbb{R}^{m \times n}$$

$\hookrightarrow$ Äußere Ableitung

$$\frac{df_3}{dz} = \text{diag}(\cos(z_1), \cos(z_2), \dots, \cos(z_m)) = \begin{pmatrix} \cos(z_1) & \dots & 0 \\ 0 & \dots & \cos(z_m) \end{pmatrix} \in \mathbb{R}^{m \times m}$$

$\hookrightarrow$ Kettenregel

$$\underline{\underline{\frac{df_3}{dx} = \frac{df_3}{dz} \cdot \frac{dz}{dx} = \text{diag}(\cos(z)) \cdot A}}$$

$\hookrightarrow$ Dimension:

$$\underline{\underline{(m \times m) \cdot (m \times n) = m \times n}}$$

2.) $f_4(z) = \tanh(z), z = Ax + b$

$\hookrightarrow A \in \mathbb{R}^{m \times n}, x \in \mathbb{R}^n, b \in \mathbb{R}^m$

$\hookrightarrow$ innere Ableitung:

$$\frac{dz}{dx} = A \in \mathbb{R}^{m \times n}$$

$\hookrightarrow$ äußere Ableitung:

$$\frac{df_4}{dz} = \text{diag}(1 - \tanh^2(z_1), \dots, 1 - \tanh^2(z_m)) \in \mathbb{R}^{m \times m}$$

$\hookrightarrow$ Kettenregel

$$\underline{\underline{\frac{df_4}{dx} = \frac{df_4}{dz} \cdot \frac{dz}{dx} = \text{diag}(1 - \tanh^2(z)) \cdot A}}$$

$\hookrightarrow$ Dimension:

$$\underline{\underline{(m \times m) \cdot (m \times n) = m \times n}}$$